

ASEN 6044**Homework #2**

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Problem 1:**Problem 2:****Problem 3:****3.1 Part A****3.2 Part B****3.3 Part C**

$$P(\rho_k | x_0) = \frac{\Gamma(\frac{\nu+1}{2})}{\Gamma(\frac{\nu}{2})\sqrt{\nu\pi}} \left(1 + \frac{(\rho_k - \sqrt{(L - \xi_0 - \dot{\xi}_0 t_k)^2 + (a_0 + \dot{a}_0 t_k - 0.5gt_k^2)})^2}{\nu}\right)^{-\frac{\nu+1}{2}} \quad (1)$$

$$P(\theta_k | x_0) = \frac{1}{\sigma_b \sqrt{2\pi}} \exp \left\{ -\frac{\left[\theta_k - \arctan\left(\frac{a_0 + \dot{a}_0 t_k - 0.5gt_k^2}{L - \xi_0 - \dot{\xi}_0 t_k} \right) \right]^2}{2\sigma_b^2} \right\} \quad (2)$$

Our negative log likelihood function is as follows:

$$\mathcal{L}(x_0; Y_{1:N}) = -\sum_{k=1}^N \log[P(\rho_k | x_0)P(\theta_k | x_0)] = J_{NLL} \quad (3)$$

3.4 Part D

In order to minimize (3), we take the gradient with respect to x_0 , or $\nabla_{x_0} \mathcal{L}(x_0; Y_{1:N})$:

$$\nabla_{x_0} \mathcal{L}(x_0; Y_{1:N}) = -\nabla_{x_0} \sum_{k=1}^N \log[P(\rho_k | x_0)P(\theta_k | x_0)]$$

It's easy to show that the expression simplifies to the following expression:

$$\begin{aligned} &= -\nabla_{x_0} \left[\frac{-\nu+1}{2} \log \left(1 + \frac{\left(\rho_k - \sqrt{(L - \xi_0 - \dot{\xi}_0 t_k)^2 + (a_0 + \dot{a}_0 t_k - 0.5gt_k^2)^2}}{\nu} \right)^2 \right) \right] \\ &\quad -\nabla_{x_0} \left[\frac{-1}{2} \left[\frac{\theta_k - \arctan\left(\frac{a_0 + \dot{a}_0 t_k - 0.5gt_k^2}{L - \xi_0 - \dot{\xi}_0 t_k} \right)}{\sigma_b} \right]^2 \right] \end{aligned} \quad (4)$$

$$\begin{aligned}
 & \nabla_{x_0} \left[\frac{-\nu+1}{2} \log \left(1 + \frac{\left(\rho_k - \sqrt{(L - \xi_0 - \dot{\xi}_0 t_k)^2 + (a_0 + \dot{a}_0 - 0.5gt_k^2)^2} \right)^2}{\nu} \right) \right] \\
 & \quad \frac{\partial}{\partial \xi_0} \left[\frac{-\nu+1}{2} \log \left(1 + \frac{\left(\rho_k - \sqrt{(L - \xi_0 - \dot{\xi}_0 t_k)^2 + (a_0 + \dot{a}_0 - 0.5gt_k^2)^2} \right)^2}{\nu} \right) \right] \\
 = & \frac{2(\rho_k - \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_0 + \dot{a}_0 t - 0.5gt_k^2)^2})(L - \xi_0 - \dot{\xi}_0 t)}{\left(\left(\rho - \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_o + \dot{a}_0 t - 0.5gt^2)^2} \right)^2 + \nu \right) \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_o + \dot{a}_0 t - 0.5gt^2)^2}} \\
 & \quad \frac{\partial}{\partial \dot{\xi}_0} \left[\frac{-\nu+1}{2} \log \left(1 + \frac{\left(\rho_k - \sqrt{(L - \xi_0 - \dot{\xi}_0 t_k)^2 + (a_0 + \dot{a}_0 - 0.5gt_k^2)^2} \right)^2}{\nu} \right) \right] \\
 = & \frac{2t(\rho_k - \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_0 + \dot{a}_0 t - 0.5gt_k^2)^2})(L - \xi_0 - \dot{\xi}_0 t)}{\left(\left(\rho - \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_o + \dot{a}_0 t - 0.5gt^2)^2} \right)^2 + \nu \right) \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_o + \dot{a}_0 t - 0.5gt^2)^2}} \\
 & \quad \frac{\partial}{\partial a_0} \left[\frac{-\nu+1}{2} \log \left(1 + \frac{\left(\rho_k - \sqrt{(L - \xi_0 - \dot{\xi}_0 t_k)^2 + (a_0 + \dot{a}_0 - 0.5gt_k^2)^2} \right)^2}{\nu} \right) \right] \\
 = & \frac{-2(\rho_k - \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_0 + \dot{a}_0 t - 0.5gt_k^2)^2})(a_0 + \dot{a}_0 t - 0.5gt^2)}{\left(\left(\rho - \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_o + \dot{a}_0 t - 0.5gt^2)^2} \right)^2 + \nu \right) \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_o + \dot{a}_0 t - 0.5gt^2)^2}} \\
 & \quad \frac{\partial}{\partial \dot{a}_0} \left[\frac{-\nu+1}{2} \log \left(1 + \frac{\left(\rho_k - \sqrt{(L - \xi_0 - \dot{\xi}_0 t_k)^2 + (a_0 + \dot{a}_0 - 0.5gt_k^2)^2} \right)^2}{\nu} \right) \right] \\
 = & \frac{-2t(\rho_k - \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_0 + \dot{a}_0 t - 0.5gt_k^2)^2})(a_0 + \dot{a}_0 t - 0.5gt^2)}{\left(\left(\rho - \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_o + \dot{a}_0 t - 0.5gt^2)^2} \right)^2 + \nu \right) \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_o + \dot{a}_0 t - 0.5gt^2)^2}}
 \end{aligned}$$

Therefore,

$$\nabla_{x_0} \left[\frac{-\nu+1}{2} \log \left(1 + \frac{\left(\rho_k - \sqrt{(L - \xi_0 - \dot{\xi}_0 t_k)^2 + (a_0 + \dot{a}_0 - 0.5gt_k^2)^2} \right)^2}{\nu} \right) \right]$$

$$= \frac{-\nu + 1}{2} \left[\frac{2(\rho_k - \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_0 + \dot{a}_0 t - 0.5gt_k^2)^2})(L - \xi_0 - \dot{\xi}_0 t)}{\left(\left(\rho - \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_0 + \dot{a}_0 t - 0.5gt^2)^2} \right)^2 + \nu \right) \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_0 + \dot{a}_0 t - 0.5gt^2)^2}} \right. \\ \left. \frac{2t(\rho_k - \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_0 + \dot{a}_0 t - 0.5gt_k^2)^2})(L - \xi_0 - \dot{\xi}_0 t)}{\left(\left(\rho - \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_0 + \dot{a}_0 t - 0.5gt^2)^2} \right)^2 + \nu \right) \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_0 + \dot{a}_0 t - 0.5gt^2)^2}} \right. \\ \left. \frac{-2(\rho_k - \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_0 + \dot{a}_0 t - 0.5gt_k^2)^2})(a_0 + \dot{a}_0 t - 0.5gt^2)}{\left(\left(\rho - \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_0 + \dot{a}_0 t - 0.5gt^2)^2} \right)^2 + \nu \right) \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_0 + \dot{a}_0 t - 0.5gt^2)^2}} \right. \\ \left. \frac{-2t(\rho_k - \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_0 + \dot{a}_0 t - 0.5gt_k^2)^2})(a_0 + \dot{a}_0 t - 0.5gt^2)}{\left(\left(\rho - \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_0 + \dot{a}_0 t - 0.5gt^2)^2} \right)^2 + \nu \right) \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_0 + \dot{a}_0 t - 0.5gt^2)^2}} \right] \quad (5)$$

$$\nabla_{x_0} \left[\frac{-1}{2} \left[\frac{\theta_k - \arctan\left(\frac{a_0 + \dot{a}_0 t_k - 0.5gt_k^2}{L - \xi_0 - \dot{\xi}_0 t_k}\right)}{\sigma_b} \right]^2 \right] \\ = \frac{-1}{2\sigma_b^2} \nabla_{x_0} \left[\theta_k - \arctan\left(\frac{a_0 + \dot{a}_0 t_k - 0.5gt_k^2}{L - \xi_0 - \dot{\xi}_0 t_k}\right) \right]^2 \\ \frac{\partial}{\partial \xi_0} \left[\theta_k - \arctan\left(\frac{a_0 + \dot{a}_0 t_k - 0.5gt_k^2}{L - \xi_0 - \dot{\xi}_0 t_k}\right) \right]^2 \\ = \frac{-2t[\theta_k - \arctan\left(\frac{a_0 + \dot{a}_0 t_k - 0.5gt_k^2}{L - \xi_0 - \dot{\xi}_0 t_k}\right)](a_0 + \dot{a}_0 t_k - 0.5gt_k^2)}{([\frac{a_0 + \dot{a}_0 t_k - 0.5gt_k^2}{L - \xi_0 - \dot{\xi}_0 t_k}]^2 + 1)(L - \xi_0 - \dot{\xi}_0 t_k)^2} \\ \frac{\partial}{\partial \dot{\xi}_0} \left[\theta_k - \arctan\left(\frac{a_0 + \dot{a}_0 t_k - 0.5gt_k^2}{L - \xi_0 - \dot{\xi}_0 t_k}\right) \right]^2 \\ = \frac{-2[\theta_k - \arctan\left(\frac{a_0 + \dot{a}_0 t_k - 0.5gt_k^2}{L - \xi_0 - \dot{\xi}_0 t_k}\right)](a_0 + \dot{a}_0 t_k - 0.5gt_k^2)}{([\frac{a_0 + \dot{a}_0 t_k - 0.5gt_k^2}{L - \xi_0 - \dot{\xi}_0 t_k}]^2 + 1)(L - \xi_0 - \dot{\xi}_0 t_k)^2} \\ \frac{\partial}{\partial a_0} \left[\theta_k - \arctan\left(\frac{a_0 + \dot{a}_0 t_k - 0.5gt_k^2}{L - \xi_0 - \dot{\xi}_0 t_k}\right) \right]^2 \\ = \frac{-2[\theta_k - \arctan\left(\frac{a_0 + \dot{a}_0 t_k - 0.5gt_k^2}{L - \xi_0 - \dot{\xi}_0 t_k}\right)]}{([\frac{a_0 + \dot{a}_0 t_k - 0.5gt_k^2}{L - \xi_0 - \dot{\xi}_0 t_k}]^2 + 1)(L - \xi_0 - \dot{\xi}_0 t_k)}$$

$$\begin{aligned}
& \frac{\partial}{\partial \dot{a}_0} \left[\theta_k - \arctan \left(\frac{a_0 + \dot{a}_0 t_k - 0.5 g t_k^2}{L - \xi_0 - \dot{\xi}_0 t_k} \right) \right]^2 \\
&= \frac{-2t[\theta_k - \arctan \left(\frac{a_0 + \dot{a}_0 t_k - 0.5 g t_k^2}{L - \xi_0 - \dot{\xi}_0 t_k} \right)]}{\left(\left[\frac{a_0 + \dot{a}_0 t_k - 0.5 g t_k^2}{L - \xi_0 - \dot{\xi}_0 t_k} \right]^2 + 1 \right) (L - \xi_0 - \dot{\xi}_0 t_k)}
\end{aligned}$$

Therefore,

$$\begin{aligned}
& \nabla_{x_0} \left[\frac{-\nu + 1}{2} \log \left(1 + \frac{\left(\rho_k - \sqrt{(L - \xi_0 - \dot{\xi}_0 t_k)^2 + (a_0 + \dot{a}_0 - 0.5 g t_k^2)^2} \right)^2}{\nu} \right) \right] \\
&= \frac{-1}{2\sigma_b^2} \left[\frac{-2t[\theta_k - \arctan \left(\frac{a_0 + \dot{a}_0 t_k - 0.5 g t_k^2}{L - \xi_0 - \dot{\xi}_0 t_k} \right)](a_0 + \dot{a}_0 t_k - 0.5 g t_k^2)}{\left(\left[\frac{a_0 + \dot{a}_0 t_k - 0.5 g t_k^2}{L - \xi_0 - \dot{\xi}_0 t_k} \right]^2 + 1 \right) (L - \xi_0 - \dot{\xi}_0 t_k)^2} \right. \\
&\quad \left. \frac{-2[\theta_k - \arctan \left(\frac{a_0 + \dot{a}_0 t_k - 0.5 g t_k^2}{L - \xi_0 - \dot{\xi}_0 t_k} \right)](a_0 + \dot{a}_0 t_k - 0.5 g t_k^2)}{\left(\left[\frac{a_0 + \dot{a}_0 t_k - 0.5 g t_k^2}{L - \xi_0 - \dot{\xi}_0 t_k} \right]^2 + 1 \right) (L - \xi_0 - \dot{\xi}_0 t_k)^2} \right. \\
&\quad \left. \frac{-2[\theta_k - \arctan \left(\frac{a_0 + \dot{a}_0 t_k - 0.5 g t_k^2}{L - \xi_0 - \dot{\xi}_0 t_k} \right)]}{\left(\left[\frac{a_0 + \dot{a}_0 t_k - 0.5 g t_k^2}{L - \xi_0 - \dot{\xi}_0 t_k} \right]^2 + 1 \right) (L - \xi_0 - \dot{\xi}_0 t_k)} \right. \\
&\quad \left. \frac{-2t[\theta_k - \arctan \left(\frac{a_0 + \dot{a}_0 t_k - 0.5 g t_k^2}{L - \xi_0 - \dot{\xi}_0 t_k} \right)]}{\left(\left[\frac{a_0 + \dot{a}_0 t_k - 0.5 g t_k^2}{L - \xi_0 - \dot{\xi}_0 t_k} \right]^2 + 1 \right) (L - \xi_0 - \dot{\xi}_0 t_k)} \right] \quad (6)
\end{aligned}$$

3.5 Part E

3.6 Part F